

LIBIN SAMKUTTY

AI Quality Engineer | Vadodara, India | samlibin2001@gmail.com | linkedin.com/in/libin-samkutty | libin-samkutty.github.io

PROFESSIONAL SUMMARY

AI Quality Engineer with 4+ years in software quality assurance including 2+ years of specialized experience evaluating large language model (LLM) systems, Retrieval-Augmented Generation (RAG) pipelines and conversational AI platforms at scale. Designed and deployed custom AI evaluation frameworks using DeepEval and RAGAS with golden datasets, multi-dimensional eval scoring and production monitoring with trace instrumentation. Deep hands-on experience testing NLP intent classification systems, semantic caching layers, vector similarity search and multi-turn conversational flows across multilingual, multi-region deployments. Bridges traditional QA rigor with AI-native evaluation methodology to deliver measurable, reproducible quality signals on non-deterministic AI systems.

CORE COMPETENCIES

LLM Evaluation, RAG Pipeline Testing, Hallucination Detection, Faithfulness Testing, DeepEval, RAGAS, Custom Eval Framework Development, Golden Dataset Design, Prompt Regression Testing, Prompt Versioning, Non-Deterministic Output Evaluation, Production Monitoring Evals, Trace Instrumentation, Model Degradation Detection, NLP Intent Classification Testing, Multi-turn Conversational Flow Testing, Semantic Cache Validation, Vector Similarity Testing, Guardrail Testing, Python, JavaScript, Pytest, Playwright, Robot Framework, REST API Automation, CI/CD, GitHub Actions, CircleCI, AWS Lambda, Docker, BigQuery, JIRA, ReportPortal, Postman

PROFESSIONAL EXPERIENCE

QA Engineer | Jeavio India Pvt. Ltd. (Client: Nivi) Jun 2023 - Present

- Led **AI quality validation** across **6 healthcare AI product lines** using DeepEval and RAGAS, defining automated evaluation frameworks for answer relevancy, faithfulness, context precision and context recall against curated **golden datasets** for a production RAG platform serving **50,000+ monthly users** across **6 countries**
- Engineered automated **production monitoring, trace instrumentation** and evaluation pipelines enabling continuous detection of **prompt regressions**, hallucinations, routing failures, intent misclassification and response quality degradation across production AI systems
- Designed test suite for **semantic caching layer**, validating that vector similarity cache hits return semantically appropriate responses, near-duplicate inputs route correctly and cache misses accurately fall through to LLM inference
- Performed systematic **RAG traffic analysis**, identifying **~26%** of pipeline invocations triggered by non-medical, vague, or social queries, delivering actionable quality signal that drove routing optimization and reduced unnecessary LLM consumption
- Designed and executed **risk-based validation strategies** for multilingual conversational AI workflows across **9 intent types** and **17 routing labels**, building targeted regression suites covering edge cases, boundary conditions and cross-intent failures
- Performed performance benchmarking and root-cause analysis across AI pipeline components, contributing to improvements in overall response time **12-18s to 4-7s**, classification **2-3s to ~900ms**, semantic cache hits **~350ms**, and RAG retrieval **8-12s to 1.5-3.5s**

Associate QA Engineer | Jeavio India Pvt. Ltd. (Client: Nivi) May 2022 - May 2023

- Authored structured test cases for NLP-driven conversational flows covering intent routing accuracy, multi-language input handling, boundary inputs and failure mode behaviour.
- Contributed to defining evaluation methodology for non-deterministic NLP outputs, establishing quality baselines in the absence of formal eval tooling and laying the groundwork for the team's first AI quality standards.
- Designed and implemented **Playwright**-based automation framework components for web and API validation, enabling **225+ automated regression scenarios**, parallel execution and CI/CD integration that reduced manual regression effort by **60%**

Project Intern | Jeavio India Pvt. Ltd. Jan 2022 - Apr 2022

- Supported manual test execution for **NLP flows** for intent classification and named entity recognition (NER) based applications, systematically documenting response inconsistencies, routing failures and edge-case behaviours
- Contributed to test case documentation for NLP routing scenarios, mapping input variations to expected intent classifications; validated REST API responses, status codes and error structures using Postman.

TECHNICAL SKILLS

Artificial Intelligence and LLM Quality Assurance: LLM Evaluation, RAG Pipeline Testing, DeepEval, RAGAS, Golden Dataset Design, Prompt Regression Testing, Hallucination Detection, Trace Instrumentation

Conversational AI Testing: Intent Classification Testing, Semantic Cache Validation, Multi-turn Flow Testing, Guardrail Testing, Adverse Input Handling, NLP Regression Testing

Testing and Automation: Pytest, Playwright, Robot Framework, Postman, REST API Automation, Schema Validation, Contract Testing, Functional, Regression, Integration

Languages and Data: Python, JavaScript, SQL; BigQuery, ETL Validation

CI/CD and Cloud: AWS Lambda, AWS EC2, Docker, GitHub Actions, CircleCI; ReportPortal, JIRA (Xray), Looker Studio

OPEN-SOURCE PROJECTS

api-test-data-generator | Stack: Python, Faker, pytest, JSON/YAML | `pip install api-test-data-generator`

- PyPI-published package for schema-driven API test data generation using JSON/YAML schemas and the Faker library.
- Supports JSON/CSV export and native pytest fixture compatibility, eliminating manual test data maintenance for API test suites.

EDUCATION

Bachelor of Engineering (B.E.), Computer Engineering | S.V.I.T., Vasad - GTU

HSC | Saint Basil School, Vadodara

SSC | Saint Basil School, Vadodara

2022 | CGPA: 9.35 / 10

2018 | 89.97 Percentile - Gujarat Board

2016 | 99.11 Percentile - Gujarat Board